

# Van Tien Pham

Researcher in Computer Science

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My research focuses on efficient AI, including pruning and low-rank approximation for neural network compression.

## Education

- 11/2025 **PhD in Computer Science**, *Université de Toulon, CNRS, LIS, UMR 7020*, France
  - Thesis: Neural network compression using pruning and low-rank approximations.
- 11/2022
- 12/2020 **Master's Degree in Computer Science**, *Hanoi University of Science and Technology (HUST)*, Vietnam
  - Thesis: Object detection, segmentation, and tracking from egocentric vision.
  - GPA: 4.0/4.0, Classification: Very good.
- 10/2019
- 6/2019 **Engineering Degree in Computer Science**, *HUST*, Vietnam
  - Programme de formation d'ingénieurs d'excellence au Vietnam.
- 9/2014 GPA: 3.2/4.0, Classification: Very good.

## Professional Experience

- 10/2022 **Software Engineer**, *Viettel Group*, Vietnam
  - Developed a driving simulator using 3D simulation engines for training purposes, including a facial recognition module for enhanced user interaction.
- 7/2019
- 2/2021 **Intern in Computer Vision**, *MICA International Research Institute, CNRS, UMI 2954*, Vietnam
  - Worked on object recognition, detection, and tracking using neural networks.
- 8/2018

## Publications

### Journals

- [J5] V.T. Pham, Y. Zniyed, T.P. Nguyen, **Coupled tensor decomposition for compact network representation**, *IEEE Transactions on Neural Networks and Learning Systems*, pp. 1-15, 2025.
- [J4] V.T. Pham, Y. Zniyed, T.P. Nguyen, **Singular values-driven automated filter pruning**, *Neural Networks*, vol. 192, p. 107857, December 2025.
- [J3] N. Tokcan\*, S.S. Sofi\*, V.T. Pham\*, C. Prévost\*, S. Kharbech\*, B. Magnier, T.P. Nguyen, Y. Zniyed, L. De Lathauwer, **Tensor decompositions for signal processing: theory, advances, and applications**, *Signal Processing*, vol. 238, p. 110191, January 2026. \*These authors contributed equally to this work as co-first authors.
- [J2] V.T. Pham, Y. Zniyed, T.P. Nguyen, **Enhanced network compression through tensor decompositions and pruning**, *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, pp. 4358–4370, March 2025.
- [J1] V.T. Pham, Y. Zniyed, T.P. Nguyen, **Efficient tensor decomposition-based filter pruning**, *Neural Networks*, vol. 178, p. 106393, October 2024.

### Preprints

- [P1] V.T. Pham, Y. Zniyed, T.P. Nguyen, **Decoupling matrix-valued functions using zeroth- and first-order information and its application in neural network compression**, to be submitted, 2025.

### Conferences

[C6] [V.T. Pham](#), Y. Zniyed, T.P. Nguyen, **Hybrid network compression through tensor decompositions and pruning**, 32nd European Signal Processing Conference (EUSIPCO), Lyon, France, pp. 1052–1056, August 2024.

[C5] [V.T. Pham](#), Y. Zniyed, T.P. Nguyen, **Élagage efficace des filtres basé sur les décompositions tensorielles**, XXIXème Colloque Francophone de Traitement du Signal et des Images (GRETSI), Grenoble, France, pp. 937-940, August 2023.

[C4] [V.T. Pham](#), T.P. Nguyen, **Identification and localization of COVID-19 abnormalities on chest radiographs**, The International Conference on Artificial Intelligence and Computer Vision, pp. 251-261, 2023.

[C3] [V.T. Pham](#), C.M. Tran, S. Zheng, T.M. Vu, S. Nath, **Chest X-ray abnormalities localization via ensemble of CNNs**, International Conference on Advanced Technologies for Communications, 2021.

[C2] [V.T. Pham](#), T.H. Tran, H. Vu, **Detection and tracking hand from FPV: benchmarks and challenges on rehabilitation exercises dataset**, IEEE International Conference on Computing and Communication Technologies (RIVF), 2021. **Best paper runner-up award**.

[C1] [V.T. Pham](#), T.L. Le, T.H. Tran, T.P. Nguyen, **Hand detection and segmentation using multimodal information from Kinect**, IEEE International Conference on Multimedia Analysis and Pattern Recognition (MAPR), 2020.

Seminars

[S2] [V.T. Pham](#), Y. Zniyed, T.P. Nguyen, **Coupled tensor decomposition for compact network representation**, Journées Apprentissage Signal Image du LIS, Châteauneuf-le-Rouge, France, November 2025.

[S1] [V.T. Pham](#), Y. Zniyed, T.P. Nguyen, **Hybrid network compression through tensor decompositions and pruning**, Journées Apprentissage Signal Image du LIS, Aix-en-Provence, France, June 2024.

Reviewer

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|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journals    | IEEE Transactions on Pattern Analysis and Machine Intelligence, Expert Systems with Applications, Neural Networks, Neurocomputing, Engineering Applications of Artificial Intelligence, Quantum Machine Intelligence, The Journal of Supercomputing, Scientific Reports, Signal, Image and Video Processing |
| Conferences | ICLR, AISTATS, ICMI, GRETSI                                                                                                                                                                                                                                                                                 |

Skills

|                      |                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|
| Programming          | Python, C, R, Pascal, $\LaTeX$ , HTML                                                |
| Frameworks and tools | PyTorch, Keras, Slurm (Jean Zay HPC), Matlab, Qt, Unreal Engine, OpenCV, Git, GitHub |
| Databases            | MySQL, PostgreSQL                                                                    |

Languages

|            |                                                       |
|------------|-------------------------------------------------------|
| English    | Professional working proficiency (Upper-Intermediate) |
| French     | Intermediate (B1)                                     |
| Vietnamese | Native proficiency                                    |

Honors

|           |                                                                  |
|-----------|------------------------------------------------------------------|
| 2016      | Honorable Mention, ACM ICPC Vietnam National Programming Contest |
| 2015      | Second Prize, Vietnam Student Olympiad in Informatics            |
| 2014–2018 | Excellence Scholarships, HUST                                    |